

THE CENTRAL QUESTION

Can IonQ capture meaningful share of a \$50B global quantum TAM by 2036 with its trapped-ion stack — or is the \$26B market cap pricing certainty Quantinuum, Google, and neutral-atom don't support?

IonQ trades at ~\$26B against \$64.7M Q1 2026 revenue and FY26 guide \$260-270M — P/S >95×. \$2.39B cash + \$470M RPO + the \$1.8B SkyWater chip-fab acquisition is the optionality. Trapped-ion gives ~99.9%+ 2Q fidelity; AQ#64 on Tempo hit three months early. But the competitive set is brutal: Quantinuum IPOs at \$12.7B with better logical-qubit performance, Google's Willow demonstrated verifiable advantage on chemistry, and neutral-atom scaled past 1,200 qubits. Honest math: cash-feasible raise schedules force heavy dilution before commercial inflection. Five scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$9.18 expected fair value today
-74.1% vs spot \$35.51

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$13.15	\$18.86	\$27.07	\$38.87
0.37× spot	0.53× spot	0.76× spot	1.09× spot

WEIGHTING RATIONALE

Ultra Bear 10% Trapped-ion scaling wall + Quantinuum dominance + commercial quantum advantage delayed past 2030; equity wipes via dilution.

Bear 30% Quantinuum's IPO + better logical qubits push IonQ to #2 within trapped-ion; smaller TAM than consensus.

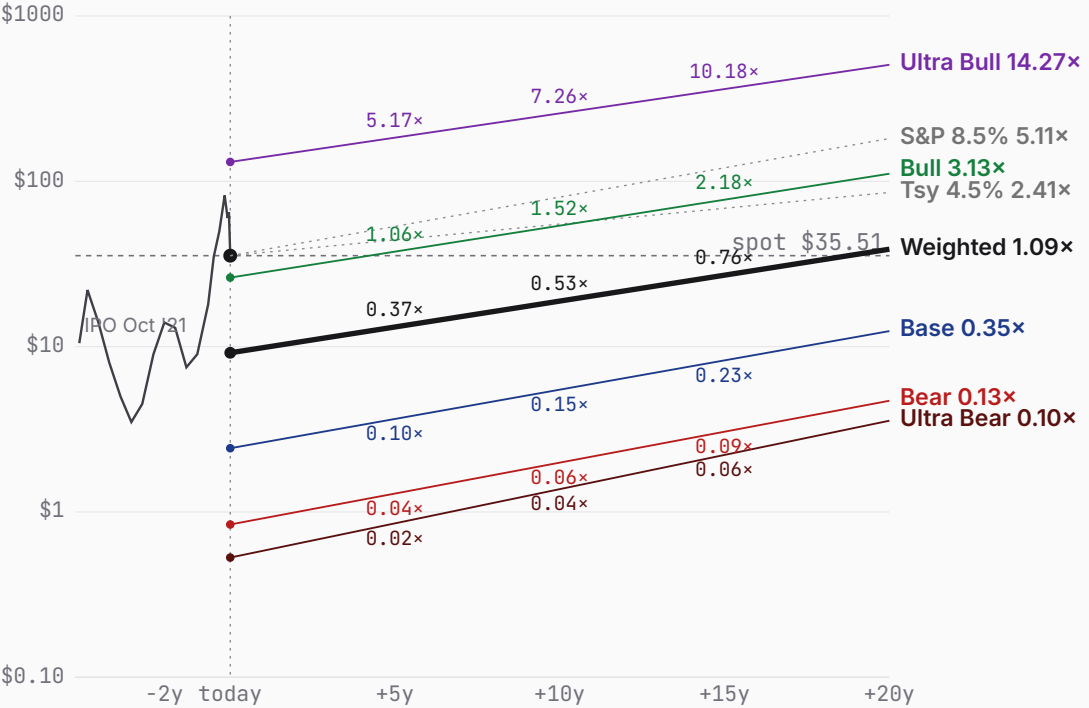
Base 42% Sequential execution on roadmap; co-leader duopoly with Quantinuum at ~8% TAM share and 17% mature margin.

Bull 15% Vertical SkyWater integration pays off; IonQ wins trapped-ion premium and platform race.

Ultra Bull 3% Tail of tails: Nvidia-of-quantum + TAM upside + competitor stumbles + multiple expansion.

Bull (15%) + Ultra Bull (3%) together contribute \$7.86 — 86% of the \$9.18 expected value despite 18% combined probability. Today's spot (\$35.51) sits 287% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$0.53	BEAR	\$0.84	BASE	\$2.43	BULL	\$26.17	ULTRA BULL	\$130.99
	-98.5% vs spot		-97.6% vs spot		-93.2% vs spot		-26.3% vs spot		+268.9% vs spot
TAM 0.7% P(fail) 60% S2C 0.6 Prob 10%		TAM 2.8% P(fail) 18% S2C 1.0 Prob 30%		TAM 7.6% P(fail) 12% S2C 1.3 Prob 42%		TAM 16.0% P(fail) 8% S2C 1.5 Prob 15%		TAM 32.0% P(fail) 4% S2C 1.7 Prob 3%	
Trapped-ion sidelined; equity wiped.		Quantinuum wins; IonQ #2 niche.		Co-leader w/ Quantinuum; ~17% mgn.		Trapped-ion wins; full-stack mgn.		Dominant quantum pure-play; software-premium re-rate.	

PROBABILITY WEIGHTS

Ultra Bear 10% Bear 30% Base 42% Bull 15% Ultra Bull 3% · Weighted expected \$9.18 (-74.1% vs spot)

ULTRA BEAR	\$0.53	BEAR	\$0.84	BASE	\$2.43	BULL	\$26.17	ULTRA BULL	\$130.99
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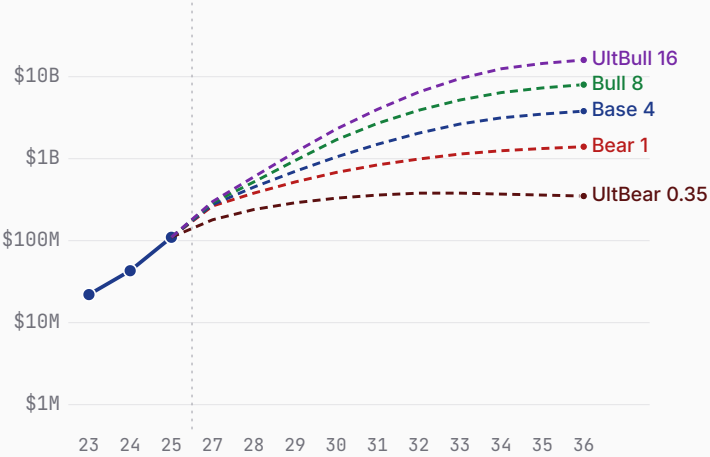
Probability 10%	Probability 30%	Probability 42%	Probability 15%	Probability 3%
Trapped-ion sidelined; equity wiped. WHY 10% 10% reflects three compounding tail risks specific to trapped-ion: fundamental scaling ceiling around 100-500 qubits — credible physicists still debate this; Quantinuum's commercial dominance within the trapped-ion category after its \$12.7B IPO; the broader risk that "useful" quantum advantage stays elusive through 2030, starving the whole sector. Each individually 25-35%; conjunctive 8-12%. SkyWater acquisition adds \$1.8B of fab opex risk if quantum demand doesn't materialize. WHAT HAPPENS Structural failure. Quantum advantage on commercial problems takes longer than the bull-deck slides project (BCG's revised "broad advantage 2030-2040" view); the NISQ era stays a science-fair through 2030. IonQ burns \$400-600M annually to AQ#256 with no commensurate revenue ramp. Worse, trapped-ion architecture hits a fundamental scaling wall around AQ#100: ion-shuttling complexity, vacuum-chamber engineering, and laser-stability budgets all degrade nonlinearly. Quantinuum's H-series, with the same physics and Honeywell's industrial-scale manufacturing, edges ahead and consolidates the trapped-ion niche. Meanwhile superconducting (IBM Nighthawk, Google Willow) achieves first verified commercial advantage at a price IonQ can't match per-shot. Neutral-atom scales past 2,000 atoms by FY28 and wins the cost curve. IonQ's \$470M RPO converts but never grows; FY30 revenue stalls below \$400M against \$500M+ cash burn including SkyWater fab opex. Cash exhausts mid-2029. Successive raises in 2028-2030 at \$40 → \$15 → \$4 dilute holders 4x. By FY32 the SkyWater fab is sold for parts; IonQ taken private at a token premium. With 60% within-scenario p_fail, distress recovery ~\$0.80. Expected per share ~\$0.50.	Quantinuum wins; IonQ #2 niche. WHY 30% Quantinuum's IPO at \$12.7B with materially better logical-qubit performance (48 logical at 99.92%) is a real competitive shift — Honeywell mfg + Cambridge Quantum SW is the most credible full-stack competitor. The broader quantum TAM consistently underperforms forecasts. IonQ executes technically but the pie is smaller and Quantinuum eats the premium slice. Within-scenario 18% p_fail reflects that even in the bear world, IonQ's cash position and DARPA contracts likely prevent bankruptcy. 30% weight reflects high-base-rate competitive entry has already happened. WHAT HAPPENS Execution exists but the market shape disappoints. IonQ delivers AQ#256 by 2027 and the SkyWater 200K-qubit chip in 2028 — real milestones. But commercial demand stays government- and research-heavy: DARPA QBI, DOE, NQI, a handful of pharma POCs. Quantinuum's deeper logical-qubit lead wins enterprise mind-share post-IPO; IonQ becomes the credible #2 in trapped-ion, capturing premium government accounts but losing the big AWS/Azure/GCP enterprise-deployment race to superconducting. Revenue ramps to ~\$1.4B FY36 — strong absolute growth but tiny vs bull-deck \$5B+. SkyWater fab is a mixed asset: useful internally but underutilized as a third-party foundry, dragging margins. Operating margin matures around 8% — better than legacy services but well below cloud-software economics the current multiple implies. Cash burn averages \$300-400M through FY29; two raises (\$400M + \$1.0B) at \$50-30 prices add ~25M shares net. Final ~440M. With 18% within-scenario p_fail, DCF/share ~\$5; expected ~\$4.20.	Co-leader w/ Quantinuum; ~17% mgn. WHY 42% Highest single bucket because the base requires no breakthrough — sequential execution on plans in motion (Tempo shipping, SkyWater vertical integration, AQ roadmap ahead of schedule, DARPA QBI participation, RPO converting). Coexistence with Quantinuum mirrors duopoly outcomes in adjacent deep-tech markets (AMD/Intel, ASML/Nikon). 42% sits below typical 50-55% base weights because quantum has more genuine tail risk than most categories — thin middle ground between "works commercially" and "doesn't." Five- scenario structure with fatter tails forces base weight down. WHAT HAPPENS Modal outcome. IonQ executes the published roadmap roughly on schedule: AQ#64 done, AQ#256 by 2027, SkyWater fab chips back in 2028, the 200K-qubit-chip class reached 2028-29. Quantinuum is a credible co-leader; the trapped-ion sub-segment becomes a duopoly (~50/50 share), with IonQ winning roughly half the US government, defense, and large-enterprise quantum-cloud business via existing AWS Braket / Azure Quantum / GCP integration. Neutral-atom takes cost-sensitive optimization; superconducting takes most "deep stack" enterprise via IBM/Google brand. By FY36 IonQ does ~\$3.8B revenue (~7.6% of \$50B TAM) at 17% mature operating margin — better than legacy hardware but below cloud-SaaS. SkyWater fab supplies internal demand plus third-party defense- electronics revenue. RPO converts steadily; multi-product customer share rises 35% → 60%+. Cash burn troughs at ~\$350M in FY28; a \$2.5B raise programme across years 1-7 at \$35-75 keeps cash≥0. SkyWater stock-portion dilution (~25M) folded in. Final shares ~450M. DCF/share ~\$16; with 12% within-scenario p_fail, expected ~\$14.70 — well below today's \$70 spot.	Trapped-ion wins; full-stack mgn. WHY 15% Chain of conditionals: AQ#256 ~on schedule (~70%), SkyWater 200K-qubit chip delivered 2028 (~60%), IonQ — not Quantinuum — captures the trapped-ion premium position (~50%), trapped-ion beats neutral-atom and superconducting on commercially relevant workloads at scale (~50%), DARPA/DOE flows weighted to IonQ (~60%). Each individually plausible; joint ~6-8% before correlation; correlated upward to ~12-18% because conditional outcomes reinforce. 15% is a real but non-modal tail. WHAT HAPPENS Trapped-ion is correct architecture and SkyWater compounds. By 2028 vertical integration via SkyWater proves out: the 200K-qubit chip arrives on schedule, error correction works, and IonQ demonstrates verifiable commercial quantum advantage on a customer workload (likely chemistry or optimization for a Fortune 50). Quantinuum stays credible #2 within trapped-ion but IonQ's vertical fab + cloud-native distribution wins the platform race. Neutral-atom struggles with fault tolerance; superconducting stalls between AQ-equivalent #500 and #2000 due to error-correction overhead. Revenue ramps to ~\$8B FY36 (~16% TAM share) with 27% mature operating margin — the SkyWater fab is now monetizing third-party customers as the dominant US trusted quantum-chip foundry. DARPA HARQ scales into a multi-billion national-security program. Federal NQI funding flows disproportionately to IonQ. Cash burn troughs FY28 then turns positive by FY30. A \$3.4B raise programme across years 1-7 at \$50-90 keeps cash≥0. Final shares ~450M. DCF/share ~\$54; with 8% within-scenario p_fail, expected ~\$50.	Dominant quantum pure-play; software-premium re-rate. WHY 3% Conjunctive bull chain × TAM upside × multiple expansion × competitor collapse. Each piece individually 40-60%; joint ~3-5%. Genuine tail capturing the world where quantum becomes a category- defining tech platform and IonQ is the dominant US pure-play. The asymmetrical-moonshots tier's reason-for-being: real upside that you can't get to without it. WHAT HAPPENS Tail of tails. Every Bull condition hits AND the FY36 TAM is materially larger than consensus (Roots Analysis \$198B FY40 trajectory, McKinsey upper-bound \$72B by 2035 just for computing) AND IonQ achieves de facto platform monopoly in trapped-ion + becomes a meaningful chip foundry via SkyWater. Quantinuum struggles with IPO-era execution; superconducting hits fault- tolerance walls that trapped-ion's higher native fidelity sidesteps. IonQ becomes the "Nvidia of quantum" — vertically integrated, full-stack, with AWS/Azure/GCP installed customer base creating a software flywheel. By FY36 IonQ does \$16B+ revenue at 33% mature operating margin (full-stack hardware + cloud + software-take-rate on partner apps). SkyWater fab serves the entire US national-security stack. Multiple expansion to software-hybrid premium. Modest \$3.1B raise in years 1-6 at \$70-100 maintains runway. Final shares ~435M. DCF/share ~\$135; with 4% p_fail, expected \$130.

IonQ business snapshot

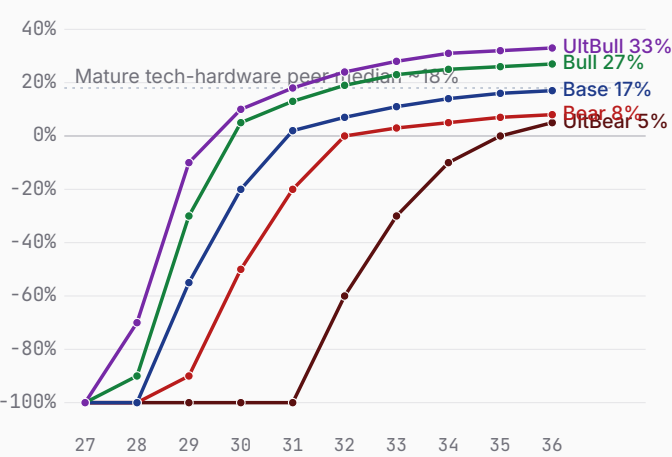
Bear \$0.84 Base \$2.43 Bull \$26.17

FY23-FY25 history + FY26 (guide) + FY27-FY36 scenario projections · fiscal years end Dec 31 · 10-K FY25, Q1 2026 10-Q, McKinsey Quantum Monitor 2026

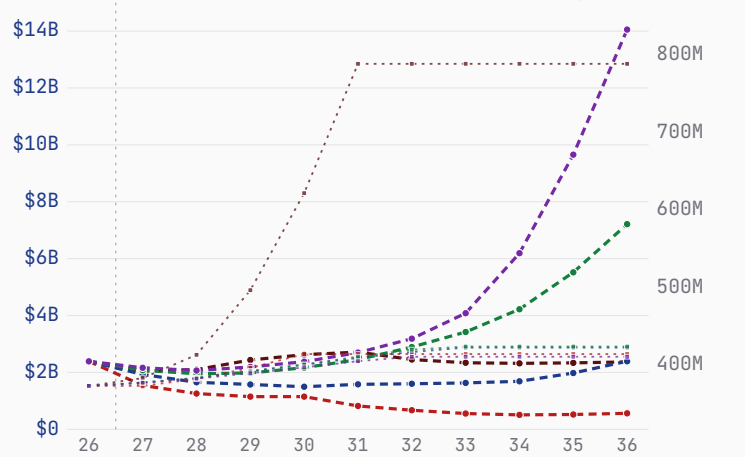
Revenue — history + scenarios to FY36 (log)



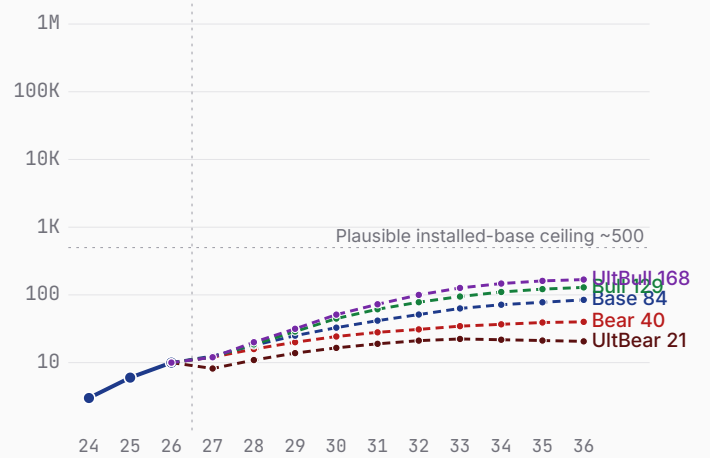
Path to profitability — op margin (FY27→FY36)



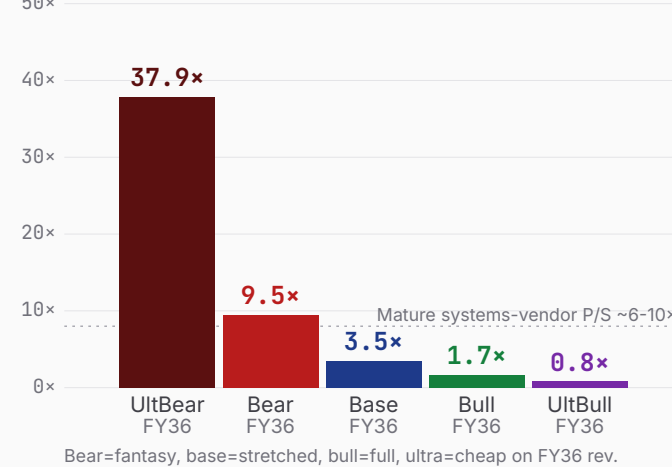
Cash + dilution (FY26 → FY36)



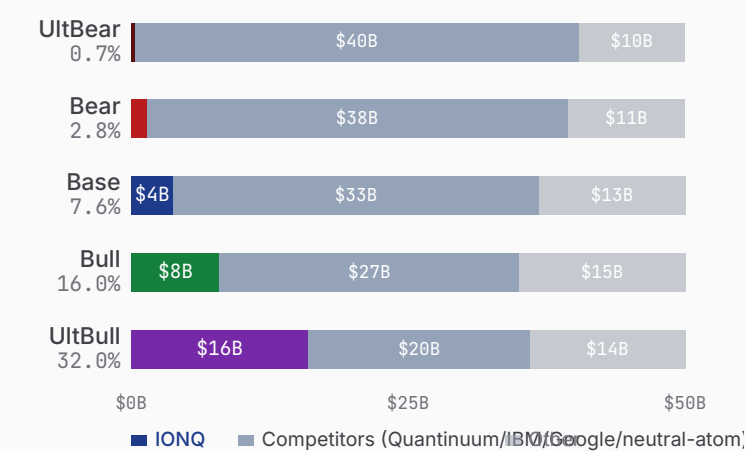
AQ#20+ class systems deployed (log)



\$13.25B mkt cap as P/S on FY36 rev



\$50B global quantum TAM (FY36) — IONQ share



Sources: IONQ Q1 2026 10-Q, FY25 10-K, IonQ Q1'26 earnings (8-K), Quantinuum S-1, McKinsey Quantum Technology Monitor 2026, BCG quantum value-creation revision, DARPA QBI Stage B selections (Nov 2025).

IonQ 7 Powers · the moat, scored

Bear \$0.84 Base \$2.43 Bull \$26.17

Arena. The race to a ~\$50B FY36 fault-tolerant quantum-computing TAM — trapped-ion (IonQ, Quantinuum) vs. superconducting (Google, IBM) vs. neutral-atom (QuEra, Pasqal, Atom Computing). **POWER ORIENTATION**
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	SkyWater fab vertical integration begun; not yet a cost-curve advantage.
Network Economies	No developer/data flywheel; cloud reach via AWS/Azure is rented, not owned.
Counter-Positioning	Trapped-ion stack incumbents can't pivot to without abandoning sunk superconducting fabs.
Switching Costs	\$470M RPO; algorithms compiled to IonQ's native gate set raise re-port cost.
Branding	Strong research credibility (AQ#64 early); not yet the enterprise default.
Cornered Resource · thesis leans here	Trapped-ion IP portfolio + SkyWater fab; Maryland/Duke talent lineage.
Process Power	Fidelity yield improving each generation; not yet a durable, opaque process edge.

COMPETITIVE READ

Real trapped-ion cornered-resource and counter-positioning leads, but network and scale power are unoriginated and Quantinuum's logical-qubit edge is the falsifier to watch; the origination window is open, but not indefinitely.

THE RACE · NAMED RIVALS

Quantinuum	\$1.4B+ raised
~20% terminal share · Helios: 48 logical qubits @ 99.92%; \$12.7B IPO; Honeywell-backed.	
Google Quantum AI	Alphabet balance sheet
~16% terminal share · Willow: verifiable quantum advantage demonstrated.	
IBM Quantum	IBM balance sheet
~14% terminal share · Nighthawk roadmap to 1,080 qubits; deep enterprise channel.	
Neutral-atom (QuEra/Pasqal/Atom)	\$300M+ aggregate
~10% terminal share · 1,200+ physical qubits in 18 months; different error profile.	

LEAD / LAG · FALSIFIERS

Algorithmic qubits (AQ)	64 vs 56	LEADING
Logical qubits @ >=99.9%	36 vs 48	LAGGING
Cash runway (yrs at burn)	4 vs 6	LAGGING

IonQ People · Opportunity · Context · Deal

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Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Niccolo de Masi1 yrs in seat

3.5%insider ownership

MEDIUMkey-person risk

Capital allocation (realized)
Niccolo de Masi became CEO in February 2025 (a director since 2021, from the SPAC-sponsor family that took IonQ public) and added the Chairman role in August 2025 when prior CEO Peter Chapman exited the board. Both academic co-founders had already left (Christopher Monroe 2023, Jungsang Kim 2024). Capital allocation is an aggressive, equity-funded acquisition roll-up: ~\$3.5B cash pro forma after a \$2B October 2025 raise, with mostly-stock deals (Oxford Ionics ~\$1.06B, SkyWater ~\$1.8B, plus Lightsynq/Qubitek/ID Quantique) driving heavy dilution. No dividend or buyback.

Incentive alignment
CEO pay is overwhelmingly equity (the exact Summary-Compensation-Table figure was not retrievable; an aggregator put it near \$90M). After ~64% say-on-pay support in 2025, the committee shifted PSU weighting toward revenue and cut maximum payouts. Insiders sold heavily in 2025 (~\$400M, including ~\$269M by the departing prior CEO and ~\$105M by de Masi); 2026 selling is smaller, mostly RSU tax withholding. Insider ownership is a derived ~3-4% (it was ~11.6% in 2024 before the share count nearly doubled).

GOVERNANCE FLAGS

- single class of common stock, one vote per share — no super-voting, no controlling shareholder (largest holders Vanguard ~10%, BlackRock ~8%)
- combined Chair & CEO (de Masi consolidated both in August 2025) with no confirmed lead independent director since early 2025 — an independent-oversight gap
- classified / staggered board (three classes) — an entrenchment feature
- both academic co-founders departed (2023, 2024) and the prior CEO fully exited (2025) — wholesale leadership turnover
- recurring say-on-pay pushback (~64% support in 2025) and ~\$400M of 2025 insider selling
- aggressive stock-funded acquisition roll-up — repeated, material dilution

PEOPLE READ

The structural positive is genuine — one share, one vote, no controlling block, so shareholders are not disenfranchised. But this is a weak 3 on practice: a combined Chair/CEO with no confirmed lead independent director, a classified board, a ~64% say-on-pay vote, ~\$400M of 2025 insider selling, and a strategy concentrated in a CEO of under two years pursuing a heavily dilutive acquisition roll-up. Key-person risk is moderate — the science survived both founders’ departures, so it is the governance concentration, not individual dependence, that is the bigger flag.

THE FOUR LEGS

Peopleobservable composite · 1–5

OpportunityThe scenario distribution · the Page-3 business snapshot

ContextThe 7 Powers analysis (Power Origination)

DealOpen-market purchase = the deal: entry price vs. the scenario distribution (the ~86.9% finding) + position sizing.

lonQ supporting analysis · disclaimers · glossary

Bear \$0.84 Base \$2.43 Bull \$26.17

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

Best-capitalized pure-play.
\$2.4B cash + \$470M RPO + SkyWater vertical integration. Multi-year runway with optionality no other public quantum has.
- 2

AQ#64 hit early.
Tempo achieved AQ#64 three months ahead of schedule — first to do so. Real execution beats slideware.
- 3

Federal tailwind is real.
DARPA QBI Stage B + HARQ + DOE/NQI's \$2.7B/5yr authorization disproportionately benefits US-based trusted-supply players.
- 4

Trapped-ion fidelity moat.
Industry-leading 2Q gate fidelity (~99.9%+) is a physics advantage, not process — hard for superconducting/neutral-atom to close.
- 5

Cloud distribution unique.
Available on AWS Braket, Azure Quantum, Google Cloud. Only Quantinuum has comparable reach; IBM/Google don't partner cross-platform.

FALSIFICATION TRIGGERS

Bear validation AQ#256 slips past 2027 · RPO conversion stalls · SkyWater close delays past Q3 2026 · Quantinuum wins a marquee enterprise displacement	Bull validation AQ#256 hit by H1 2027 · First verified commercial quantum advantage on customer workload · Multi-product share >50% · 2nd large govt contract (>\$100M)	Reframe needed If neutral-atom (QuEra/Atom Computing) achieves fault-tolerance first — trapped-ion premium thesis erodes; revisit TAM share assumptions across all scenarios
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Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$IONQ or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Trapped-ion qubits — lonQ's qubit modality — ytterbium ions in electromagnetic traps, manipulated by lasers. Highest gate fidelity but slower clock than superconducting.	AQ (Algorithmic Qubits) — lonQ's benchmark for usable capacity. AQ#N = circuits requiring 2^N states. AQ#64 hit 2026; AQ#256 targeted 2027.	Quantum advantage — Quantum solving a problem faster/cheaper than classical. Google's Willow demonstrated verifiable advantage on chemistry Oct 2025; no commercial advantage verified yet.
Quantinuum — Honeywell-backed full-stack trapped-ion competitor. Filed for \$12.7B IPO May 2026; better logical-qubit performance (48 logical at 99.92%).	SkyWater — US semiconductor foundry lonQ is acquiring (~\$1.8B, closes Q2/Q3 2026). Domestic chip fab — enables 200K-qubit chip in 2028.	Tempo / Forte — lonQ system generations. Forte = current commercial (AQ#36). Tempo = next-gen, AQ#64 demonstrated, 256-qubit chip.